

OUTRAM PARK

Current capabilities, and where we are going

Theodore Ong

National University of Singapore

September 14, 2026

Open-source Unified TRAnsient Multi-Physics Advanced Reactor simulation Kit

What OUTRAM PARK is

A single Cargo workspace housing a pure-Rust reactor simulation suite.

Scale (measured 14 Sep 2026)

- **37** member crates
- **1.06 M** lines under `src/`
- **3,104** Rust source files
- **8,367** test functions
- GPL-3.0 throughout
(one AGPL-3.0 exception: KOVAN)

Design constraints

- Enums for dispatch — no dyn, no Box
- No lifetime parameters; `Arc<RwLock<T>>` for shared state
- uom-typed physical quantities
- Android/Termux *and* wasm32 are supported targets
- Release-mode builds and tests only

Not a wrapper. Every upstream code listed later is an independent Rust translation, not an FFI binding.

The honest frame: verification is not validation

- **Verification** — is it implemented correctly?
- **Validation** — does it represent physical reality well enough for its purpose?

Every crate carries a two-axis maintainer sign-off tracker. Today, both axes read “not yet manually checked” on every crate — AI-assisted output is treated as untrusted draft material until a human reviews it.

Prototype → Unit Tested → Integrated → **Verified** → Validated → Published

Intended for education, research, capability building and V&V only. Explicitly not for facility operation, reactor control, licensing decisions, safety-critical analysis or emergency response.

Nine of 37 crates are declared mature

Crate	Bar it is held to	Evidence
tampines-steam-tables	IF97 region eqns to 10^{-8} vs IAPWS [19]	reference standard
tuas_boussinesq_solver	CIET outlet temp within 0.2 °C; Gnielinski 2% [6]	experimental + cross-code
outram-foam-appbuilder-lib	Sod Table II vs exact Riemann; L_2 within 5% of peak [17]	analytical / MMS
chem-eng-...-simulator	discretisations exact at samples vs closed form	analytical
outram-foam-basic-lib	conservation to 10^{-12} ; convergence order matches theory	analytical / MMS
njoy-outram-park-fork	agrees with NJOY2016 to 7 significant figures [11]	cross-code
outram-mc-libs	k_{eff} within 500 pcm of ICSBEP Godiva [12]	cross-code
teh-o-prke	published β reproduced; PRKE limiting cases exact	unit + consistency
outram-park-fork-liggghts	integrator + contact laws vs closed form [8]	analytical / MMS

Published-benchmark agreement is deliberately not an entry requirement — the validation pipeline runs through these APIs, so demanding it first would deadlock the work.

Neutronics and nuclear data

`njoy-outram-park-fork` — 94k lines

all nuclear data lives here; transport crates are data-free

- RECONR / BROADR / THERMR / ACER ported from NJOY2016 [11]
- Faddeeva kernel, windowed multipole
- Lean-ACE and WMP data blobs, $\bar{\nu}$ / χ
- LEAPR regenerates graphite $S(\alpha, \beta)$ from a 12 kB deck

`outram-mc-libs` — 41k lines

- CSG geometry, particle tracking
- k -eigenvalue; delta (Woodcock) tracking for doubly heterogeneous media
- Depletion via CRAM [14]
- Optional wgpu compute, CPU fallback mandatory

Code-to-code against OpenMC [16] is live — k -eff, six-factor, ring RPT — at **70%** of its tracked scope.

Thermal-hydraulics, CFD and process

Crate	Lines	Capability
tuas_boussinesq	157k	Boussinesq single-phase TH; CIET coupled-DRACS natural circulation
tampines-steam-tables	138k	IAPWS-IF97 [19] + steam-turbine equations + choked flow
...fork-dwsim-libs	91k	Process-simulation blocks; rigorous + dynamic distillation [10]
...fork-coolprop	67k	Helmholtz-EOS properties, 137 fluids [2]
outram-foam-basic-lib	58k	OpenFOAM primitive + FV layer, Layers 1–4 [20]
outram-foam-appbuilder	41k	Solver-application layer; hosts the GeN-Foam port
...fork-offbeat	60k	Fuel performance: eigenstrain, creep, gap/contact, ~70 correlations
tampines	—	Framework composing TUAS, CoolProp, steam tables, FV layer

... abbreviates the *outram-park-* prefix.

Plus mesh generation — blockMesh, snappyHexMesh and a cfMesh port — a Blender-derived surface-authoring frontend at **94%**, and turbulence closures with $k-\omega$ SST implemented.

Structural mechanics — farrer-park

Finite-element Analysis for Reactor Reliability, Engineering Response, Plasticity And Risk. Ported from MOOSE [5] and PRISMS.

Verified against analytical solutions

- MMS convergence orders match theory
- Patch test at machine precision
- Newton convergence order **2.004**
- Crystal plasticity vs analytic Schmid yield: **1.5e-15** relative
- Frame indifference **7.6e-16**

Volumetric locking: measured and cured

- Full integration: MMS order 0.63–1.24
- B-bar: **2.00–2.05**
- Plastic collapse vs closed-form limit load: +23.9% → +**1.06%**

*Shear locking is measured and **not** cured — 11.25% too stiff at two Quad4 through the depth, 66.7% at aspect ratio 4. Small strain only: no lattice reorientation, no backstress. Verification only; no human V&V.*

Supporting layers

KOVAN — deterministic knowledge layer

- Literature archive with access tiers and provenance
- rust-analyzer-backed symbol queries
- Graph digitiser — calibration and uncertainty are structurally mandatory
- Offline, no cloud, no vector store

RAFFLES — UQ and risk [1]

- Distributions, samplers, Sobol sensitivity
- Surrogates: not yet implemented

Digital twin engine — 57k lines

- egui simulators: HTR-10, FHR
- OPC-UA server and client
- Offline demonstrations only — never connected to plant systems

MSRE trio [15] — depletion [13], salt thermochemistry, circulating-fuel neutronics [9] *All three are AI-assisted drafts with no human V&V, and none is wired into a simulator yet.*

What is *not* working — stated deliberately

- **916 open issues**, 674 closed. Four P0 defects are todo! () panics or arithmetic underflow in live solver paths.
- **No crate has human V&V sign-off.** Every bookkeeping tracker reads INCOMPLETE.
- **The first headless run of `htgr_sim_v1`** (6 Sep 2026) showed power overshooting to **27.8 MW**, $\sim 2.8\times$ nominal, against a docstring claiming it started near steady state. The docstring had been wrong, unnoticed, because a GUI-only simulator cannot be tested.
- **IF97 flash panics above 1073.15 K** — too low for an HTGR.
- **`wasm32` gate proves compilation, not execution.** `thread::spawn` and `Instant::now` compile and panic at runtime.
- **Six extern "C" calls** to the system libm remain, duplicated across three crates.

These are tracked, not hidden. A simulator whose behaviour cannot be falsified is not a capability.

Near-term: consolidation

PETIR — core maths crate

Polynomials, Equations, Transforms, Integration, Roots

- no_std port of GSL [4], Octave and SLATEC [3]
- **Ported, never AI-generated** — traceable to a library with a real test suite
- Removes the last C from the workspace
- Bit-identical transcendentals across platforms
- Unblocks NJOY and RAFFLES, neither of which can take the CFD crate

Why it matters [18]

- Chebyshev fitting for cross-section representation
- Orthogonal-basis regression for surrogate models
- ~27k lines of general numerics currently sit inside an OpenFOAM-branded crate

Also near-term

- Android/Termux build sweep
- API dogfooding: *if it is too complex for a small model, it is a bad API*

Medium-term: reactor targets

Programme	Done	Scope
OpenMC code-to-code	70%	<i>k</i> -eff, six-factor, ring RPT [16]
boon-lay TRISO release	73%	Lagrangian decay / fission-product transport
NJOY module coverage	22%	ERRORR, GROUPR, URR, covariance
HTR-10 pebble bed	0%	Retarget the simulator from prismatic to pebble [7]
HTGR roadmap (HTTR + HTR-10)	0%	Real-time capable prismatic and pebble branches
CIET educational simulator v3	0%	TAMPINES library + digital-twin GUI
Nuclear oil refinery	0%	HTR-10 process heat into a refinery flowsheet

Longer term — BISHAN, Building Internal Source-term and Hazard Analysis Network: Level 2 PSA. Severe-accident progression, containment response, in-building aerosol transport and pool scrubbing, release categories. Depends on RAFFLES for the probabilistic machinery.

The through-line

What is genuinely built

- Nuclear data agreeing with NJOY2016 to 7 figures
- MC within 500 pcm of Godiva
- TH validated against CIET to 0.2 °C
- FEM with MMS orders matching theory
- A million lines under one licence, one dependency policy, one workspace

What makes it trustworthy

- Every port names its upstream, commit and licence
- Every benchmark states methodology *and* measured result
- Every gap is a tracked issue, not a silence
- AI-written code is untrusted until a human signs it off

The hard part is not writing the physics.

It is being able to prove, later, that it was right.

References I

-  Andrea Alfonsi, Cristian Rabiti, Diego Mandelli, Joshua Cogliati, Robert Kinoshita, and Antonio Naviglio.
RAVEN and dynamic probabilistic risk assessment: Software overview.
Proceedings of ESREL, 2014.
-  Ian H. Bell, Jorrit Wronski, Sylvain Quoilin, and Vincent Lemort.
Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp.
Industrial & Engineering Chemistry Research, 53(6):2498–2508, 2014.
-  Kirby W. Fong, Thomas H. Jefferson, Tokihiko Suyehiro, and Lee Walton.
SLATEC common mathematical library, version 4.1.
Technical report, Sandia National Laboratories, Los Alamos National Laboratory, Air Force Weapons Laboratory, 1993.

References II

-  Mark Galassi, Jim Davies, James Theiler, Brian Gough, Gerard Jungman, Patrick Alken, Michael Booth, and Fabrice Rossi.
GNU Scientific Library Reference Manual.
Network Theory Ltd., 3 edition, 2009.
-  Derek Gaston, Chris Newman, Glen Hansen, and Damien Lebrun-Grandié.
MOOSE: A parallel computational framework for coupled systems of nonlinear equations.
Nuclear Engineering and Design, 239(10):1768–1778, 2009.
-  Volker Gnielinski.
New equations for heat and mass transfer in turbulent pipe and channel flow.
International Chemical Engineering, 16(2):359–368, 1976.
-  International Atomic Energy Agency.
Evaluation of high temperature gas cooled reactor performance: Benchmark analysis related to initial testing of the HTTR and HTR-10.
Technical Report IAEA-TECDOC-1382, IAEA, 2003.

References III

-  Christoph Kloss, Christoph Goniva, Alice Hager, Stefan Amberger, and Stefan Pirker.
Models, algorithms and validation for opensource DEM and CFD-DEM.
Progress in Computational Fluid Dynamics, 12(2/3):140–152, 2012.
-  Alexander Lindsay, Gavin Ridley, Andrei Rykhlevskii, and Kathryn Huff.
Introduction to Moltres: An application for simulation of molten salt reactors.
Annals of Nuclear Energy, 114:530–540, 2018.
-  William L. Luyben.
Distillation Design and Control Using Aspen Simulation.
Wiley, 2 edition, 2013.
-  R. E. Macfarlane, D. W. Muir, R. M. Boicourt, A. C. Kahler, and J. L. Conlin.
The NJOY nuclear data processing system, version 2016.
Technical Report LA-UR-17-20093, Los Alamos National Laboratory, 2017.

References IV

-  Nuclear Energy Agency.
International handbook of evaluated criticality safety benchmark experiments.
Technical Report NEA/NSC/DOC(95)03, OECD Nuclear Energy Agency, 2020.
-  M. H. A. Piro, S. Simunovic, T. M. Besmann, B. J. Lewis, and W. T. Thompson.
The Thermochemica software package for modeling multi-component thermodynamic equilibria.
Journal of Nuclear Materials, 414(3):399–407, 2011.
-  Maria Pusa.
Rational approximations to the matrix exponential in burnup calculations.
Nuclear Science and Engineering, 169(2):155–167, 2011.
-  R. C. Robertson.
MSRE design and operations report, part I: Description of reactor design.
Technical Report ORNL-TM-728, Oak Ridge National Laboratory, 1965.

References V

 Paul K. Romano, Nicholas E. Horelik, Bryan R. Herman, Adam G. Nelson, Benoit Forget, and Kord Smith.

OpenMC: A state-of-the-art Monte Carlo code for research and development.

Annals of Nuclear Energy, 82:90–97, 2015.

 Gary A. Sod.

A survey of several finite difference methods for systems of nonlinear hyperbolic conservation laws.

Journal of Computational Physics, 27(1):1–31, 1978.

 Lloyd N. Trefethen.

Approximation Theory and Approximation Practice.

SIAM, 2013.

 Wolfgang Wagner and Hans-Joachim Kretzschmar.

International Steam Tables: Properties of Water and Steam Based on the Industrial Formulation IAPWS-IF97.

Springer, 2 edition, 2008.

References VI

-  H. G. Weller, G. Tabor, H. Jasak, and C. Fureby.
A tensorial approach to computational continuum mechanics using object-oriented techniques.
Computers in Physics, 12(6):620–631, 1998.